

Preprint: Challenges in Adopting Microservices Architecture: A Systematic Review of Data Consistency and Fault Tolerance (Draft Version) (Submitted)

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Preprint Version Information

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Abstract:	Microservices architecture has gained significant attention as a dominant paradigm for building scalable and cloud-native applications by decomposing monolithic systems into independently deployable services with decentralized data ownership. However, this architectural approach introduces challenges related to distributed data management and system reliability. This paper presents a systematic literature review examining data consistency and fault tolerance mechanisms in microservices environments. The study analyzes research published between 2016 and 2026 collected from major academic databases including IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, Google Scholar, and arXiv. The findings indicate that strict consistency models often limit system scalability and availability, leading many architectures to adopt eventual consistency and BASE principles. Saga-based transaction management patterns are increasingly preferred over traditional Two-Phase Commit protocols due to improved resilience, although they introduce additional implementation complexity. The review also highlights the lack of standardized evaluation frameworks for benchmarking distributed resilience strategies. Overall, the study emphasizes the importance of balancing consistency, scalability, and fault tolerance when designing reliable microservices-based systems.
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